AP (19.2) (34) 10.16. 2026

The Purkinje fibers have a fast ~~inherit~~ inherent conduction rate, & the electrical impulses reaches all of the ventricular muscle cells in about 75 ms

Since the electrical stimulus begins at the Apex, the contraction also begins at the apex & travels toward the base of the heart, similar to ~~se~~ squeezing a tube of toothpaste from the bottom.

This allows, the blood to be pumped out of the ventricles & into the ~~aro~~ aorta & pulmonary trunk.

The total time elapsed from the initiation of the impulse in the SA node until depolarization of the ventricles is approx 225 ms

Membranes Potential & Ion Movement in Cardiac Cells

Action potentials are considerably different between cardiac conductive cells & contractive cells → while Na^+ & K^+ -> are essential → So is Ca^{2+} → also critical.

Skeletal muscles & neurons have [stable resting potential]

Cardiac conductive cells do NOT

cc cells contain a series of Na^{2+} ion channels that allow a normal & slow influx of Na^{2+} → causing membrane potential to rise slowly from an initial value of -60 mV up to about -40 mV

→ this resulting movement of Na^{2+} → creates ~~sop~~ spontaneous depolarization (prepotential depolarization)